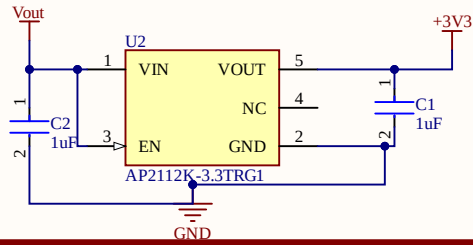
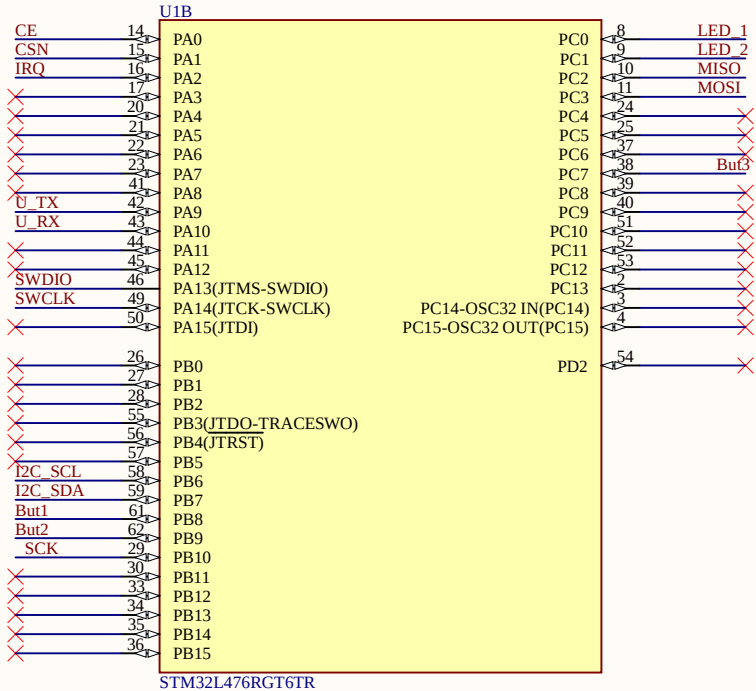


POWER SUPPLY



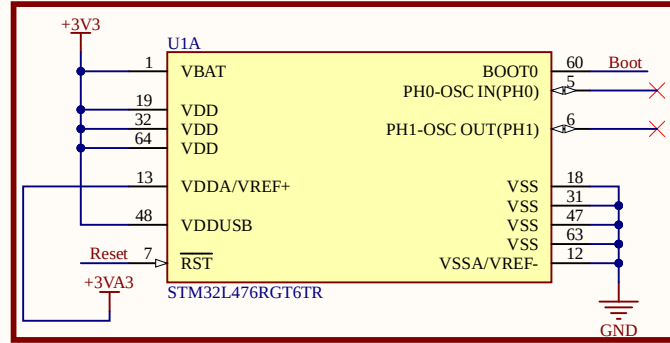
MCU



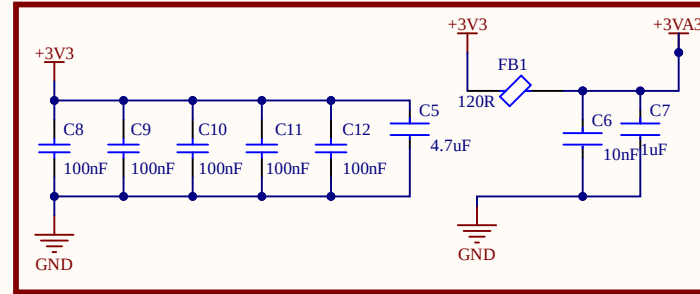
MpuHand

Made for Practice purpose,by Bakhtiyar

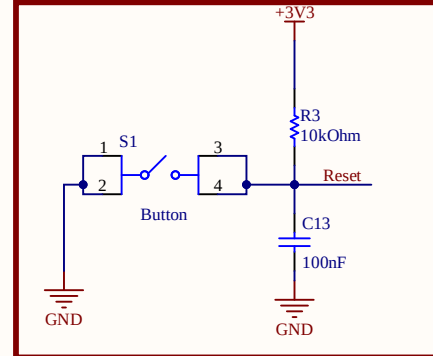
Has to be connected to bypass cap



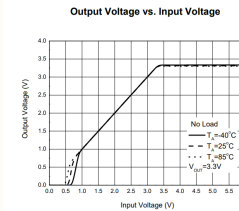
Bypass Cap



Reset



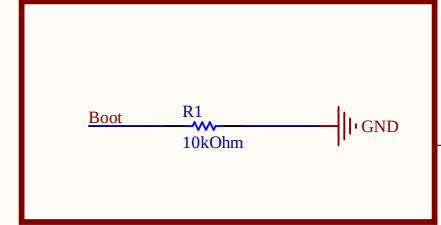
Led requires 20 mA
 $I = U/R$
 $> R = (3.3 - 2.1) / 20\text{mA}$ So we need approx $> 60\text{Ohm}$



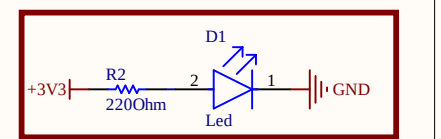
MP3 Player

Size	Number	Revision
A4		
Date:	6.01.2026	Sheet of
File:	C:\Users\...\mpustm.SchDoc	Drawn By:

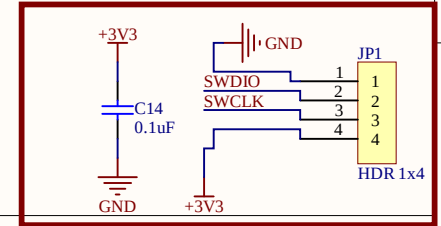
Boot

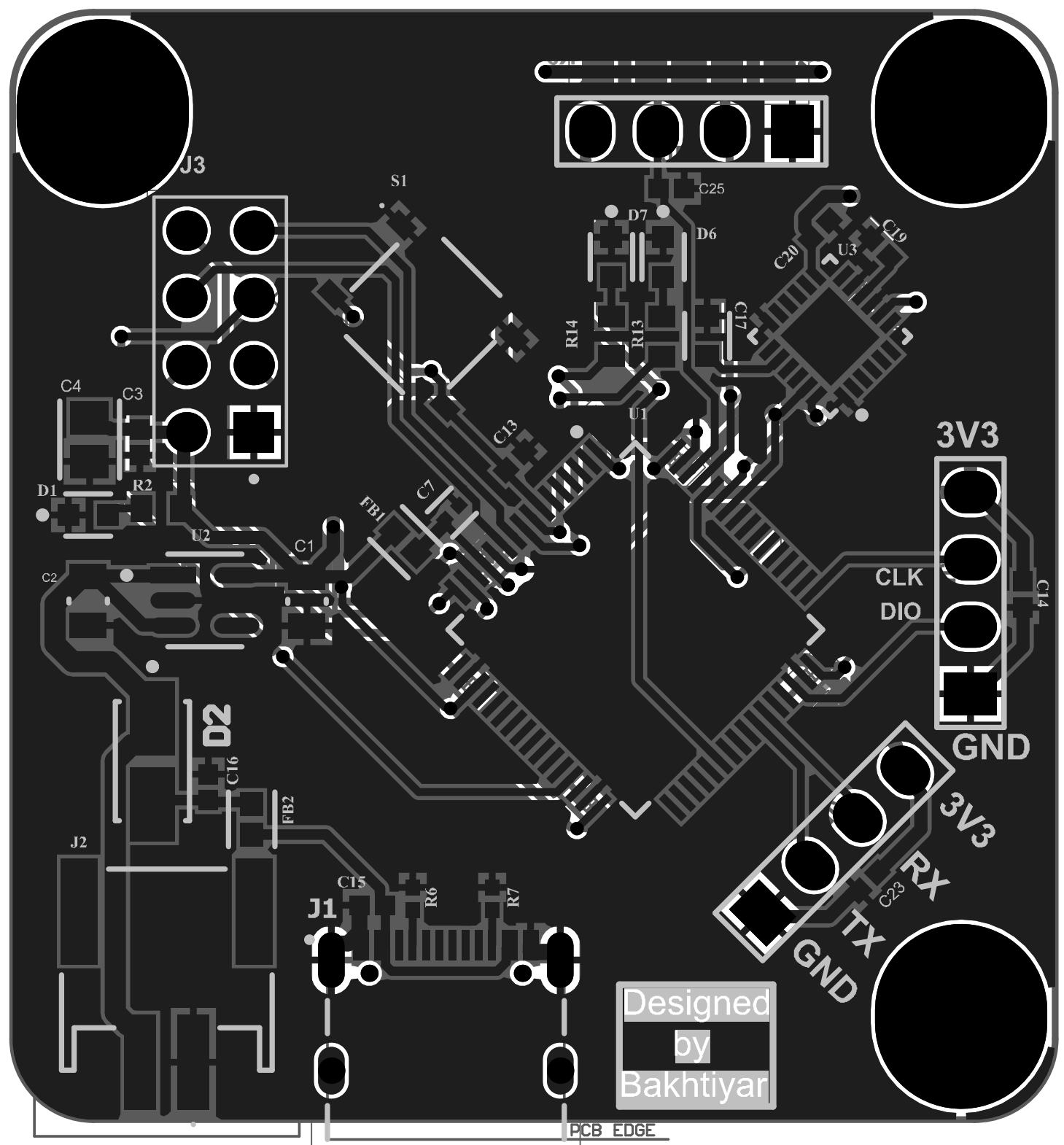


Power indicator



JTAG





Comment	Description	Designator	Footprint	LibRef	Quantity
1uF	CAP CER1UF 50V X7R 0805	C1, C2	FP-CC0805-0_85-MFG	OMP-03422-000864-1	2
100nF	Chip Capacitor, 100 nF, +/- 10%, 16 V, -55 to 125 degC, 0402 (1005 Metric), RoHS, Tape and Reel	C3, C8, C9, C10, C11, C12, C13, C15, C16, C24, C25, C27	CAPC1050X55X23ML0 5T13	OMP-2000-05439-1	12
10uF	CL21 Series 0805 10uF 16V±10% Tolerance X5R Multilayer Ceramic Chip Capacitor	C4	FP-CL21-IPC_B	OMP-2000-06102-2	1
4.7uF	Cap Ceramic 4.7uF 10V X5R±20% SMD 0402 +85°C Paper T/R	C5	FP-CL282-IPC_C	OMP-13271-000548-1	1
10nF	Cap Ceramic 10nF 50V X7R±10% SMD 0402 +125°C Paper T/R	C6	FP-CL165-IPC_C	OMP-13271-000387-1	1
1uF	Cap Ceramic 1uF 10V X5R±10% SMD 0402 +85°C Paper T/R	C7	FP-CL511-IPC_C	OMP-13271-001936-1	1
0.1uF	Chip Capacitor, 100 nF, +/- 10%, 16 V, -55 to 125 degC, 0402 (1005 Metric), RoHS, Tape and Reel	C14, C19, C20, C23	CAPC1050X55X23ML0 5T13	OMP-2000-05439-1	4
2.2nF	CAP CER 2200PF 50V X7R 0603	C17	FP-CC0603-DA-MFG	OMP-03422-000445-1	1
10nF	CAP CER 10000PF 50V X7R 0402	C18	FP-CC0402-MFG	OMP-2100-03592-2	1
LED	LED GREEN CLEAR CHIP SMD	D1, D6, D7	FP-LTST-C191GKT-MFG	OMP-22017-000029-1	3
SS34	Transient Voltage Suppressor, 6.40 to 231 V, 7.5 V Stand-off Voltage, Uni-Directional, -55 to 155 degC, 2-pin DO-214AC, Pb-Free, Tape and Reel	D2	VISH-DO-214AC2_M	OMP-0544-00028-2	1
120R	Chip Ferrite Bead, 0603, 50 @ 100MHz, 0.05Ω, 25%, 800mA	FB1, FB2	FP-BUM18-0_15-10_8-IPC_C	OMP-0686-00425-3	2
USB4105GFA060	USB-C (USB TYPE-C) USB 2.0 Receptacle Connector 24 (16+8 Dummy) Position Surface Mount, Right Angle, Through Hole	J1	GCT_USB4105GFA060	USB4105GFA060	1
PH-2PMB	Connector	J2	PH-2PMB	PH-2PMB	1
DS1021-2X4SF162-B	Pin header, pin strips; male, PIN: 8; straight; 2.54mm; THT; 2x4	J3	CONNFLY_DS1021-2X4SF162-B	DS1021-2X4SF162-B	1
HDR 1x4	CONN HEADER VERT 4POS 2.54MM	JP1, JP2, JP3	HDR_Male1_4 2.54mm	68001-204HUF	3
10kOhm	Chip Resistor, 10 kOhm, +/- 5%, 0.125 W, -55 to 155 degC, 0805 (2012 Metric), RoHS, Tape and Reel	R1, R3, R8, R9, R10	RESC2013X60X35ML10 T20	OMP-2000-05025-1	5
220Ohm	General Purpose Chip Resistors, 220 Ohm, +/- 1% -55 to 155 degC, 0603 (1608 Metric), RoHS, Tape and Reel	R2, R13, R14	RESC1608X55X25ML10 T15	OMP-1659-00018-2	3
4k7Ohm		R4, R5	FP-RCD402-0_4-MFG	OMP-2002-08011-2	2
5k1Ohm		R6, R7	RESC1005X40X25ML05 T10	OMP-2002-07753-1	2
Button	Switch (4-Pin Push)	SL	GSPTS004901	GSPTS004901	1
STM32L476RGT6TR	IC MCU 32BIT 1MB FLASH 64LQFP	U1	FP-LQFP64-5W-MFG	OMP-12105-000039-1	1
AP2112K-3.3TRG1	IC REG LINEAR 3.3V 600MA SOT25	U2	FP-SOT25-IPC_A	OMP-12756-000035-1	1
MPU-6050	3-Axis Gyroscope, Accelerometer and a Digital Motion Processor(TM) (DMP), 2.375 to 3.46 V, -40 to 85 degC, 24-Pin QFN, RoHS, Tape and Reel	U3	ISQFN-24_V	OMP-1753-00002-2	1